Oracle ORA Error Commands Runbook

OS Level, Database Level, and Infrastructure Level Diagnostics for Common ORA Codes

Scope: Linux/UNIX-focused Oracle Database operations, including single-instance, Oracle Restart, RAC, ASM, Listener, and Data Guard checks. Examples align mainly with Oracle Database 19c+ practices and official Oracle references.

Important: This is a diagnostic and triage reference, not a replacement for your site runbook. Commands that change state, delete files, resize storage, kill sessions, or restart resources are intentionally marked as review-before-use. Always capture evidence first.

Contents

- 1. How to use this runbook
- 2. Universal ORA triage sequence
- 3. ORA code range map
- 4. OS-level command checklist and ORA matrix
- 5. Database-level command checklist and ORA matrix
- 6. Infrastructure-level command checklist and ORA matrix
- 7. Copy/paste command snippets
- 8. Escalation evidence checklist
- 9. Official reference links

1. How to use this runbook

1. Start from the exact error stack, not only the first ORA code. Many Oracle errors include a primary ORA plus secondary ORA, TNS, Linux errno, ASM, RMAN, or CRS messages.
2. Classify the first actionable layer: OS, database, or infrastructure. If unsure, run the universal triage commands and the quick map below.
3. Capture alert log, trace, listener log, CRS/ASM logs, and SQL evidence before making changes. This protects RCA quality and helps with Oracle Support escalation.
4. Use the matrix rows as first-response checks. The same ORA code can appear from several root causes, so confirm using evidence before applying a fix.

Safety rule: For production, do not run ALTER SYSTEM, ALTER DATABASE, RMAN DELETE, srvctl stop/start, crsctl stop/start, or kill-session commands until impact, backup/restore status, and change approval are clear.

2. Universal ORA triage sequence

Universal triage flow

Step	Action	Command idea / evidence
1	Record exact stack	Copy the full client/server error stack including secondary ORA/TNS/CRS/RMAN/Linux errno messages and timestamp.
2	Identify environment	Capture ORACLE_SID, ORACLE_HOME, TNS_ADMIN, host, OS user, database version, instance status, and RAC node.
3	Lookup official meaning	Use oerr when available, then Oracle Error Help for the exact code and release. Use first ORA arguments for ORA-00600/07445.
4	Locate ADR and alert log	Query V$DIAG_INFO and use ADRCI SHOW ALERT/SHOW INCIDENT/SHOW TRACEFILE.
5	Check layer-specific evidence	OS: filesystem/memory/process/network. DB: instance/session/space/undo/SGA/FRA. Infra: listener/RAC/ASM/Data Guard.
6	Choose least risky action	Prefer reversible/online actions first: register service, add space, clear expired metadata, fix path/permission, add disk, or adjust service config according to policy.
7	Document result	Keep before/after commands, timestamps, SQL IDs, incident IDs, and any files changed.

3. ORA code range map

Quick map by ORA family

Ranges are only a guide. Oracle code ranges overlap; use the exact error documentation and the surrounding stack.

ORA family	Typical area and examples
ORA-00000 to ORA-00999	SQL execution, locking, internal kernel, or parse/schema issues. Examples: ORA-00054, ORA-00060, ORA-00600, ORA-00942.
ORA-01000 to ORA-01499	Session, login, cursor, data, or PL/SQL call behavior. Examples: ORA-01017, ORA-01031, ORA-01400.
ORA-01500 to ORA-01699	Storage, segment, undo/temp, and recovery/file access. Examples: ORA-01555, ORA-01652, ORA-01653.

ORA family	Typical area and examples
ORA-01900 to ORA-02099	Users, privileges, quotas, profiles, and distributed transactions. Examples: ORA-01950, ORA-02049.
ORA-03100 to ORA-03199	Session/network disconnect symptoms. Examples: ORA-03113, ORA-03135.
ORA-04000 to ORA-04099	PL/SQL, library cache, shared pool, and package/trigger errors. Examples: ORA-04031, ORA-04088.
ORA-09900 to ORA-09999	Operating system or audit/file creation failures. Example: ORA-09925.
ORA-12154 / ORA-125xx	Oracle Net naming, listener, and transport errors. Examples: ORA-12154, ORA-12514, ORA-12541.
ORA-15000 to ORA-15199	Oracle ASM disk group, disk, file, and directory errors. Examples: ORA-15041, ORA-15063.
ORA-19800 to ORA-19999	Fast Recovery Area, RMAN, flashback/recovery space. Examples: ORA-19809, ORA-19815.
ORA-27000 to ORA-27999	OS-dependent I/O, memory, and system-call errors. Examples: ORA-27041, ORA-27102, ORA-27300.
ORA-29700 to ORA-29799	RAC and Cluster Synchronization Services. Example: ORA-29701.

4. OS-level command checklist and ORA matrix

- Use OS checks when the ORA stack includes ORA-27xxx, ORA-099xx, Linux/UNIX errno, file path errors, memory allocation failures, audit file errors, or process startup problems.
- Always check the Oracle OS owner context. Many problems are caused by correct commands run from the wrong OS account, wrong ORACLE_HOME, or wrong ORACLE_SID.

```
# Fast OS baseline
id; groups; hostname -f; date; uname -a
echo ORACLE_SID=$ORACLE_SID ORACLE_HOME=$ORACLE_HOME TNS_ADMIN=$TNS_ADMIN
ps -ef | egrep 'pmon|tnslsnr|asm_pmon|ocssd|crsd|ohasd' | grep -v grep
df -hT; df -ih
free -h; vmstat 1 5; iostat -xz 1 5
ulimit -a; ipcs -m
dmesg -T | tail -200
```

OS-level ORA error-to-command matrix

ORA code(s)	Likely meaning / first clue	First commands to run
ORA-00845	MEMORY_TARGET or AMM cannot allocate memory; on Linux this commonly points to /dev/shm size or support. [S12]	df -h /dev/shm mount | grep shm sqlplus / as sysdba -> show parameter memory adrci exec="show alert -tail 100"
ORA-01034 + ORA-27101	Oracle not available / shared memory realm does not exist. Often database is down, wrong ORACLE_SID, or PMON not running.	echo $ORACLE_SID $ORACLE_HOME ps -ef | grep pmon | grep -v grep sqlplus / as sysdba -> startup adrci exec="show alert -tail 100"

Oracle ORA Error Command Reference - OS / DB / Infrastructure

ORA code(s)	Likely meaning / first clue	First commands to run
ORA-01078	Failure processing system parameters. Usually bad PFILE/SPFILE value, missing file, wrong path, or startup parameter conflict.	ls -l $ORACLE_HOME/dbs/init*.ora $ORACLE_HOME/dbs/spfile*.ora sqlplus / as sysdba -> create pfile='/tmp/init.ora' from spfile grep -i "ORA-01078" $ORACLE_BASE/diag/*/*/*/trace/alert*.log
ORA-01157 + ORA-01110	Database cannot identify or lock a data file. Can be OS path, storage, ASM alias, permission, or missing mount point.	sqlplus / as sysdba -> select file#,name,status from v$datafile; ls -l <datafile_path> asmcmd ls +DATA/<db_unique_name>/DATAFILE dmesg -T | tail -100
ORA-09925	Unable to create audit trail file. Oracle Error Help says to check the UNIX error number for an OS error. [S11]	sqlplus / as sysdba -> show parameter audit_file_dest ls -ld <audit_file_dest> df -h <audit_file_dest> touch <audit_file_dest>/write_test && rm <audit_file_dest>/write_test
ORA-12547	TNS lost contact during local or remote connection. Common OS checks: oracle binary permissions, libraries, listener, and environment.	ls -l $ORACLE_HOME/bin/oracle ldd $ORACLE_HOME/bin/oracle | grep "not found" lsnrctl status echo $ORACLE_HOME $ORACLE_SID $LD_LIBRARY_PATH
ORA-19502 / ORA-27072	Write error or file I/O error during backup, archiving, or datafile operation. Check filesystem, ASM, NFS, SAN, and kernel logs.	df -hT; df -ih iostat -xz 1 5 dmesg -T | egrep -i "error|fail|scsi|multipath" asmcmd lsdg adrci exec="show alert -tail 200"
ORA-19504 / ORA-27040	Failed to create file or OS create-file failure. Usually path, permission, file exists, quota, or destination full.	ls -ld <directory> df -h <directory> getfacl <directory> 2>/dev/null umask su - oracle -c "touch <directory>/write_test"
ORA-27041 / ORA-27037	OS could not open or stat a file. Oracle Error Help for ORA-27041 says to check errno. [S7]	ls -l <file> stat <file> namei -l <file> mount | grep <mount> dmesg -T | tail -100
ORA-27102 / ORA-27125	Out of memory or shared memory segment creation failure; check memory, kernel params, HugePages, ulimits. [S8]	free -h ulimit -a sysctl kernel.shmmax kernel.shmall kernel.sem fs.aio-max-nr cat /proc/meminfo | egrep "Huge|Mem" ipcs -m
ORA-27300 / ORA-27301 / ORA-27302	OS dependent operation failed. Treat the additional OS error text as the first clue.	adrci exec="show alert -tail 200" grep -i "ORA-273" <trace_file> dmesg -T | tail -200 journalctl -xe --no-pager | tail -200
ORA-04030	Process private memory allocation failure. Often OS memory pressure, PGA pressure, process limits, or bad SQL workload.	free -h; vmstat 1 5 ulimit -a sqlplus / as sysdba -> select * from v$pgastat; ps -eo pid,ppid,user,rss,vsz,cmd --sort=-rss | head

Prepared as a field runbook. Validate fixes against change control and backup policy before execution.

ORA code(s)	Likely meaning / first clue	First commands to run
ORA-00600 / ORA-07445	Internal error or exception. ADR normally creates incidents and trace files. Collect evidence before restart or cleanup. [S6][S14]	adrci exec="show incident -mode detail" adrci exec="show problem" adrci exec="show tracefile -rt" grep -Ei "ORA-00600|ORA-07445" alert_${ORACLE_SID}.log

5. Database-level command checklist and ORA matrix

- Use DB-level checks for SQL, schema, user/security, sessions/locks, temp/undo/tablespace, SGA/PGA, package/trigger, and internal incident errors.
- For CDB/PDB systems, confirm the correct container. Many object, privilege, quota, and service issues are container-specific.

```
sqlplus / as sysdba
set lines 220 pages 1000
select instance_name, status, database_status, host_name from v$instance;
select name, open_mode, log_mode, database_role from v$database;
select con_id, name, open_mode from v$pdbs order by con_id;
select name, value from v$diag_info order by name;
```

Database-level ORA error-to-command matrix

ORA code(s)	Likely meaning / first clue	First commands to run
ORA-00001	Unique constraint violated. Find constraint and duplicate key source.	select owner, constraint_name, table_name, index_name from dba_constraints where constraint_name = upper('<constraint>'); select cols.table_name, cols.column_name from dba_cons_columns cols where cols.constraint_name = upper('<constraint>');
ORA-00054	Resource busy NOWAIT or timeout expired. Identify blockers and locked objects.	select sid,serial#,username,blocking_session,event,sql_id from v$session where blocking_session is not null; select l.session_id,o.owner,o.object_name,o.object_type,l.locked_mode from v$locked_object l join dba_objects o on o.object_id=l.object_id;
ORA-00060	Deadlock detected. Use trace/alert details to identify participating SQL and objects.	adrci exec="show alert -tail 200" adrci exec="show tracefile -rt" select sid,serial#,username,event,sql_id,blocking_session from v$session where wait_class <> 'Idle';

ORA-00904 / ORA-00942	Invalid identifier or table/view does not exist. Validate schema, object name, synonyms, grants, and case-sensitive identifiers.	select owner,object_name,object_type,status from dba_objects where object_name = upper('<object>');
		select * from dba_tab_privs where table_name = upper('<object>');
		select * from dba_synonyms where synonym_name = upper('<object>');

ORA-01000	Maximum open cursors exceeded. Identify sessions and SQL with high cursor use.	select s.sid,s.serial#,s.username,count(*) open_cur from v$open_cursor oc join v$session s on s.sid=oc.sid group by s.sid,s.serial#,s.username order by open_cur desc; show parameter open_cursors

ORA-01017	Invalid credential or not authorized; logon denied. Oracle Error Help lists invalid credentials/authorization as the cause. [S2]	sqlplus -L <user>/<password>@<connect_identifier>
		select username,account_status,lock_date,expiry_date from dba_users where username=upper('<user>');
		select * from dba_role_privs where grantee=upper('<user>');

Oracle ORA Error Command Reference - OS / DB / Infrastructure

ORA code(s)	Likely meaning / first clue	First commands to run
ORA-01031	Insufficient privileges. Check object privileges, role usage, definer/invoker rights, and OS authentication for SYSDBA/SYSASM.	select * from session_privs order by privilege; select * from dba_sys_privs where grantee=upper('<user>'); id; groups; grep -E "dba|asm|oper" /etc/group
ORA-01400 / ORA-12899	Data issue: cannot insert NULL or value too large. Confirm column definition and incoming data length.	desc <owner>.<table> select column_name, data_type, data_length, nullable from dba_tab_columns where owner=upper('<owner>') and table_name=upper('<table>');
ORA-01403 / ORA-01422	PL/SQL SELECT INTO returned no rows or too many rows. Validate query predicates and expected cardinality.	select sql_id,sql_text from v$sql where sql_text like '%<key text>%'; -- Re-run exact SELECT with predicates and count(*) before code change.
ORA-01555	Snapshot too old. Investigate undo retention, long queries, undo pressure, and commit frequency.	show parameter undo select begin_time,end_time,undoblks,txncount,maxquerylen,tuned_undoretention from v$undostat order by begin_time desc fetch first 24 rows only; select sid,serial#,sql_id,program,logon_time from v$session where status='ACTIVE';
ORA-01652	Unable to grow temp or other segment. Oracle Error Help recommends adding/resizing space or autoextend where appropriate. [S1]	select * from v$tempseg_usage; select tablespace_name,file_name,bytes/1024/1024 mb,autoextensible,maxbytes/1024/1024 max_mb from dba_temp_files; select tablespace_name,sum(bytes)/1024/1024 free_mb from dba_free_space group by tablespace_name;
ORA-01653 / ORA-01654	Unable to extend table or index. Check tablespace free space, autoextend, max size, and segment owner.	select tablespace_name,file_name,bytes/1024/1024 mb,autoextensible,maxbytes/1024/1024 max_mb from dba_data_files where tablespace_name=upper('<tablespace>'); select owner,segment_name,segment_type,bytes/1024/1024 mb from dba_segments where tablespace_name=upper('<tablespace>') order by bytes desc;
ORA-01950	No privileges or quota on tablespace. Check user quotas and default tablespace.	select username,default_tablespace from dba_users where username=upper('<user>'); select * from dba_ts_quotas where username=upper('<user>');
ORA-02049	Distributed transaction lock timeout. Check database links, remote sessions, and in-doubt transactions.	select * from dba_2pc_pending; select owner,db_link,username,host from dba_db_links; select sid,serial#,event,blocking_session,sql_id from v$session where event like '%dblink%' or blocking_session is not null;
ORA-04031	Out of shared memory. Oracle Error Help points to insufficient shared memory/pool sizing. [S5]	show parameter sga select pool,name,bytes/1024/1024 mb from v$sgastat order by bytes desc fetch first 30 rows only; select component,current_size/1024/1024 mb from v$sga_dynamic_components order by current_size desc;
ORA-06502 / ORA-06512	PL/SQL numeric/value error or stack line reference. Use line number plus object source to find failing assignment/cast.	select owner,name,type,line,text from dba_source where owner=upper('<owner>') and name=upper('<package_or_proc>') and line between <line>-5 and <line>+5 order by line;
ORA-28000 / ORA-28001	Account locked or password expired. Check profile and account status before reset/unlock.	select username,account_status,lock_date,expiry_date,profile from dba_users where username=upper('<user>'); select * from dba_profiles where profile=(select profile from dba_users where username=upper('<user>'));
ORA-00600 / ORA-07445	Internal error/exception. Do not guess based only on code; use first argument, trace, incident, patch level, and Oracle Support notes. [S6][S14]	adrci exec="show incident -mode detail" adrci exec="ips create package incident <incident_id>" select banner_full from v$version; select comp_id,version,status from dba_registry;

Prepared as a field runbook. Validate fixes against change control and backup policy before execution.

6. Infrastructure-level command checklist and ORA matrix

- Use infrastructure checks for Oracle Net, listener, SCAN/VIP/DNS/firewall, RAC/Clusterware, ASM, FRA/archiving, RMAN destinations, and Data Guard transport/apply errors.
- When using Oracle Restart/RAC, prefer SRVCTL for Oracle-managed resources. Use CRSCTL for Clusterware stack checks and non-Oracle resources unless Oracle Support/site policy says otherwise.

```
# Listener / network
lsnrctl status; lsnrctl services; tnsping <alias>
ss -ltnp | grep 1521; getent hosts <host>; nc -vz <host> <port>

# RAC / Restart / ASM
srvctl status database -db <db_unique_name> -detail
crsctl check crs; crsctl check cluster -all; crsctl stat res -t
asmcmd lsdg; asmcmd lsdsk -k
```

Infrastructure-level ORA error-to-command matrix

ORA code(s)	Likely meaning / first clue	First commands to run
ORA-12154	Connect identifier could not be resolved. Oracle Error Help recommends checking connection string, sqlnet.ora naming methods, tnsnames.ora location/access/syntax, LDAP, or Easy Connect. [S3]	echo $TNS_ADMIN tnsping <alias> cat $TNS_ADMIN/tnsnames.ora cat $TNS_ADMIN/sqlnet.ora sqlplus -L user/pass@//host:port/service
ORA-12514	Listener does not know the requested service. Oracle Error Help recommends lsnrctl services and checking database/PDB/service registration. [S4]	lsnrctl services [listener_name] lsnrctl status [listener_name] sqlplus / as sysdba -> show parameter service_names sqlplus / as sysdba -> alter system register;
ORA-12541	No listener on target host/port. Validate listener process, port, firewall, and address.	ps -ef | grep tnslsnr | grep -v grep lsnrctl status ss -ltnp | grep 1521 nc -vz <host> <port> firewall-cmd --list-all 2>/dev/null
ORA-12545	Target host or object does not exist. Usually hostname, DNS, SCAN, VIP, or routing issue.	getent hosts <host> nslookup <host> ping -c 3 <host> traceroute <host> ip route get <ip>
ORA-12560	Protocol adapter error. On Windows often service/SID; on Linux often environment/listener/local bequeath checks.	echo $ORACLE_HOME $ORACLE_SID $PATH sqlplus / as sysdba lsnrctl status ps -ef | grep pmon | grep -v grep
ORA-03113 / ORA-03135	Connection ended or lost contact. Symptom, not root cause: check alert log, network, server process trace, firewall, and resource pressure.	adrci exec="show alert -tail 200" adrci exec="show tracefile -rt" ss -tanp | grep <port> dmesg -T | tail -200 lsnrctl status

Oracle ORA Error Command Reference - OS / DB / Infrastructure

ORA code(s)	Likely meaning / first clue	First commands to run
ORA-00257	Archiver error. Oracle Error Help says archiver could not archive redo; common cause is destination full or mandatory destination failed. [S13]	archive log list select dest_id,status,error,destination from v$archive_dest_status; select name,space_limit/1024/1024 limit_mb,space_used/1024/1024 used_mb,space_reclaimable/1024/1024 reclaim_mb from v$recovery_file_dest; df -h <archive_dest>
ORA-19809 / ORA-19815	FRA limit exceeded or warning. Check recovery file destination usage before deleting anything.	show parameter db_recovery_file_dest select * from v$recovery_file_dest; select file_type,percent_space_used,percent_space_reclaimable,number_of_files from v$flash_recovery_area_usage; rman target / -> report obsolete;
ORA-16014 / ORA-16038	Redo archival/log transport failure. Check archive destinations, FRA, remote standby, and listener services.	select dest_id,status,error,destination,target from v$archive_dest_status order by dest_id; select sequence#,first_time,applied from v$archived_log order by sequence# desc fetch first 20 rows only; dgmgrl / -> show configuration
ORA-15041	ASM disk group space exhausted. Oracle Error Help recommends adding disks, deleting files, or completing rebalance if imbalanced. [S10]	asmcmd lsdg sqlplus / as sysasm -> select name,state,total_mb,free_mb,usable_file_mb from v$asm_diskgroup; asmcmd du +DATA/<db_unique_name> asmcmd ls +DATA
ORA-15032 / ORA-15063 / ORA-15025	ASM operation failed, insufficient disks discovered, or disk cannot be opened. Check discovery string, permissions, multipath/udev, ASM alert log.	sqlplus / as sysasm -> show parameter asm_diskstring sqlplus / as sysasm -> select path,header_status,mount_status,state,total_mb,free_mb from v$asm_disk; ls -l /dev/mapper/* multipath -ll adrci exec="show alert -tail 200"
ORA-15173	ASM alias/file path not found. Validate disk group, directory, and alias spelling.	asmcmd pwd asmcmd ls -l +DATA/<path> asmcmd find +DATA "*<filename>*" sqlplus / as sysasm -> select name from v$asm_alias where name like '%<name>%';
ORA-29701 / ORA-29702	Cluster Synchronization Service or cluster group issue. Oracle Error Help for ORA-29701 says verify CSS state and traces. [S9]	crsctl check crs crsctl check cluster -all crsctl stat res -t ps -ef | egrep "ocssd|crsd|ohasd" | grep -v grep adrci exec="show homes"
ORA-29740 / node eviction symptoms	RAC node eviction or split-brain protection event. Check CSS, interconnect, voting disks, and time sync logs.	crsctl query css votedisk olsnodes -n -s -t cluvfy comp clocksync -n all dmesg -T | tail -200 adrci exec="show alert -tail 300"
Data Guard ORA-166xx / ORA-167xx	Broker or redo apply/log transport issue. Treat with DG configuration, alert logs, and standby status.	dgmgrl / -> show configuration dgmgrl / -> show database verbose <db_unique_name> select process,status,thread#,sequence#,block# from v$managed_standby; select dest_id,status,error from v$archive_dest_status;

Prepared as a field runbook. Validate fixes against change control and backup policy before execution.

7. Copy/paste command snippets

Universal ORA lookup and evidence collection

```
# OS-level lookup if Oracle message files are installed
oerr ora 01652

# Locate diagnostic directories from SQL
sqlplus / as sysdba
set lines 220 pages 1000
select name, value from v$diag_info order by name;

# Read latest alert log with ADRCI
adrci exec="show homes; show alert -tail 100"
adrci exec="show incident -mode detail"
adrci exec="show tracefile -rt"
```

Linux/UNIX operating system baseline

```
id; groups; hostname -f; date; uname -a
echo ORACLE_SID=$ORACLE_SID ORACLE_HOME=$ORACLE_HOME TNS_ADMIN=$TNS_ADMIN
ps -ef | egrep 'pmon|tnslsnr|asm_pmon|ocssd|crsd|ohasd' | grep -v grep
df -hT; df -ih
free -h; vmstat 1 5
iostat -xz 1 5
ulimit -a
sysctl kernel.shmmax kernel.shmall kernel.sem fs.aio-max-nr 2>/dev/null
dmesg -T | tail -200
```

Database health and status SQL

```
sqlplus / as sysdba
set lines 220 pages 1000
select instance_name, status, database_status, host_name from v$instance;
select name, open_mode, log_mode, database_role from v$database;
select con_id, name, open_mode from v$pdbs order by con_id;
select resource_name,current_utilization,max_utilization,limit_value
from v$resource_limit
where resource_name in ('sessions','processes','open_cursors');
```

Tablespace, temp, undo, and FRA checks

```
select tablespace_name, round(sum(bytes)/1024/1024) free_mb
from dba_free_space group by tablespace_name order by free_mb;

select tablespace_name, file_name, round(bytes/1024/1024) mb,
    autoextensible, round(maxbytes/1024/1024) max_mb
```

```
from dba_data_files order by tablespace_name, file_name;

select username, tablespace, sql_id, blocks*8192/1024/1024 used_mb
from v$tempseg_usage order by used_mb desc;

select name, space_limit/1024/1024 limit_mb, space_used/1024/1024 used_mb,
    space_reclaimable/1024/1024 reclaimable_mb
from v$recovery_file_dest;
```

Locks, sessions, and SQL evidence

```
select sid, serial#, username, status, event, blocking_session, sql_id
from v$session
where blocking_session is not null or wait_class <> 'Idle'
order by blocking_session nulls last, sid;

select l.session_id, o.owner, o.object_name, o.object_type, l.locked_mode
from v$locked_object l join dba_objects o on o.object_id = l.object_id
order by l.session_id;

select sql_id, child_number, executions, elapsed_time/1000000 elapsed_sec, sql_text
from v$sql
where sql_id = '<sql_id>';
```

Listener and Oracle Net checks

```
echo TNS_ADMIN=$TNS_ADMIN
lsnrctl status [listener_name]
lsnrctl services [listener_name]
lsnrctl version
tnsping <alias>
sqlplus -L user/password@//host:port/service_name
ss -ltnp | grep 1521
getent hosts <host>
```

RAC, Clusterware, and Oracle Restart checks

```
srvctl status database -db <db_unique_name> -detail
srvctl config database -db <db_unique_name>
srvctl status service -db <db_unique_name>
srvctl status listener
crsctl check crs
crsctl check cluster -all
crsctl stat res -t
crsctl query css votedisk
```

ASM checks

```
asmcmd lsdg
asmcmd lsdsk -k
asmcmd ls +DATA
sqlplus / as sysasm
select name,state,type,total_mb,free_mb,usable_file_mb from v$asm_diskgroup;
select path,header_status,mount_status,state,total_mb,free_mb from v$asm_disk order by path;
```

RMAN and archive cleanup evidence - review before delete

```
rman target /
LIST ARCHIVELOG ALL;
CROSSCHECK ARCHIVELOG ALL;
REPORT OBSOLETE;
# Only after confirming retention and backup policy:
# DELETE EXPIRED ARCHIVELOG ALL;
# DELETE OBSOLETE;
```

8. Escalation evidence checklist

- Exact error stack, client command, timestamp with timezone, database name/DB_UNIQUE_NAME, instance/SID, host/node, and PDB/service name.
- Alert log covering at least 30 minutes before and after the event; for RAC, collect all relevant nodes and ASM homes.
- ADRCI incident/problem details, trace files, and IPS package for ORA-00600, ORA-07445, ORA-04031, corruption, or repeated instance failures.
- Listener status/services output, listener log excerpt, tnsnames.ora/sqlnet.ora excerpt, and connection string for ORA-12154/125xx issues.
- Filesystem, ASM disk group, FRA, archive destinations, and RMAN output for space, backup, archiver, or file I/O errors.
- SRVCTL/CRSCTL output for RAC/Restart/Clusterware issues: crsctl check crs, crsctl check cluster -all, crsctl stat res -t, ocrcheck, and voting disk status as applicable.
- Recent change history: patching, storage/network change, password/profile change, service relocation, parameter modification, OS reboot, or certificate/wallet change.

High-risk command examples that require review

Review-before-use commands

Command family	Risk and evidence needed
alter system kill session	Can roll back work, disrupt applications, or mask root cause. Capture blocker SQL/session evidence first.
alter database datafile/tempfile resize/add/drop	Can affect storage layout and recoverability. Confirm ASM/filesystem space, backup, and tablespace type.
rman delete obsolete / delete archivelog	Can break recoverability or standby shipping if retention/apply status is misunderstood.
srvctl stop/start database/service/listener	Can cause outage or failover. Check service policy, drain timeout, and application impact.
crsctl stop crs/cluster	Can stop many resources on a node/cluster. Use only under approved maintenance or emergency procedure.

Command family	Risk and evidence needed
manual rm of Oracle files	Can destroy data, archives, backups, wallets, or diagnostics. Use Oracle tools where possible and preserve evidence.

9. Official reference links

Source tags in the matrix point to these official Oracle references. The command syntax and behavior can differ by release, platform, and Grid/Database home, so verify against your installed version.

[S1] Oracle Database Error Help - ORA-01652. https://docs.oracle.com/en/error-help/db/ora-01652/

[S2] Oracle Database Error Help - ORA-01017. https://docs.oracle.com/en/error-help/db/ora-01017/

[S3] Oracle Database Error Help - ORA-12154. https://docs.oracle.com/en/error-help/db/ora-12154/

[S4] Oracle Database Error Help - ORA-12514. https://docs.oracle.com/en/error-help/db/ora-12514/

[S5] Oracle Database Error Help - ORA-04031. https://docs.oracle.com/en/error-help/db/ora-04031/

[S6] Oracle Database Error Help - ORA-00600. https://docs.oracle.com/en/error-help/db/ora-00600/

[S7] Oracle Database Error Help - ORA-27041. https://docs.oracle.com/en/error-help/db/ora-27041/

[S8] Oracle Database Error Help - ORA-27102. https://docs.oracle.com/en/error-help/db/ora-27102/

[S9] Oracle Database Error Help - ORA-29701. https://docs.oracle.com/en/error-help/db/ora-29701/

[S10] Oracle Database Error Help - ORA-15041. https://docs.oracle.com/en/error-help/db/ora-15041/

[S11] Oracle Database Error Help - ORA-09925. https://docs.oracle.com/en/error-help/db/ora-09925/

[S12] Oracle Database Error Help - ORA-00845. https://docs.oracle.com/en/error-help/db/ora-00845/

[S13] Oracle Database Error Help - ORA-00257. https://docs.oracle.com/en/error-help/db/ora-00257/

[S14] Oracle Database Utilities 19c - ADRCI. https://docs.oracle.com/en/database/oracle/oracle-database/19/sutil/oracle-adr-command-interpreter-adrci.html

[S15] Oracle Database Reference 19c - V$DIAG_INFO. https://docs.oracle.com/en/database/oracle/oracle-database/19/refrn/V-DIAG_INFO.html

[S16] Oracle Net Services Reference 19c - Listener Control Utility. https://docs.oracle.com/en/database/oracle/oracle-database/19/netrf/listener-control-utility.html

[S17] Oracle RAC Administration 19c - SRVCTL Command Reference. https://docs.oracle.com/en/database/oracle/oracle-database/19/racad/server-control-utility-reference.html

[S18] Oracle Clusterware Administration 19c - CRSCTL Utility Reference. https://docs.oracle.com/en/database/oracle/oracle-database/19/cwadd/oracle-clusterware-control-crsctl-utility-reference.html

[S19] Oracle Automatic Storage Management 19c - About ASMCMD. https://docs.oracle.com/en/database/oracle/oracle-database/19/ostmg/about-asmcmd.html

[S20] Oracle Database Administrator Guide 19c - Starting Up and Shutting Down. https://docs.oracle.com/en/database/oracle/oracle-database/19/admin/starting-up-and-shutting-down.html

[S21] Oracle SQL*Plus Commands 19c. https://docs.oracle.com/en/database/oracle/oracle-database/19/sqlqr/SQL-Plus-Commands.html

Release note: This document intentionally focuses on common, high-value triage patterns rather than every ORA code. For any code not listed, use the same sequence: exact stack -> Oracle Error Help/oerr -> alert/trace -> layer-specific checks -> evidence-backed action.